Android Studio Installation

Step-by-step guide for setting up Flutter development on Windows 10/11

In this guide

- Check your computer requirements
- Download and install Android Studio
- Install Android SDK and command-line tools
- Set up an Android emulator
- Install Flutter and connect it to Android Studio
- Run flutter doctor and fix common issues

Official downloads

Android Studio: https://developer.android.com/studio

Flutter install guide: https://docs.flutter.dev/get-started/install/windows

Java / Android build setup: Android Studio includes a suitable JDK for typical Android builds; you generally do not need to install Java separately.

1. Before installation

• Use a supported 64-bit Windows 10 or Windows 11 computer.

• Make sure you have a stable internet connection and enough free disk space for Android Studio, SDK packages and emulator images. The exact space required varies by components.

• For a faster emulator, hardware virtualization should be enabled in BIOS/UEFI and supported by your computer.

• If you are setting up Flutter, download the Flutter SDK too; Android Studio alone does not install Flutter itself.

2. Download Android Studio

• Open the official Android Studio page: https://developer.android.com/studio

• Select the Windows download and read/accept the license terms.

• Save the installer, then open the downloaded .exe file.

3. Install Android Studio

• Run the installer. If Windows asks for permission, choose Yes only if you downloaded it from the official Android Developers website.

• In the setup wizard, keep Android Studio selected. Keep Android Virtual Device selected if offered and you plan to use an emulator.

• Choose an installation location or accept the default, then select Install.

• When installation finishes, launch Android Studio.

• On first launch, choose Do not import settings if this is your first installation.

• Choose the Standard setup for a typical beginner installation. Review the components and accept the SDK license agreements.

• Wait for the SDK and related components to download, then click Finish.

4. Check Android SDK components

From the Android Studio welcome screen, open More Actions → SDK Manager. If a project is already open, use Tools → SDK Manager.

In the SDK Platforms tab, select a current stable Android SDK platform. In SDK Tools, ensure these are installed (names can vary slightly by version):

• Android SDK Build-Tools

• Android SDK Platform-Tools (includes adb)

• Android SDK Command-line Tools (latest)

• Android Emulator (if you want to use a virtual device)

Click Apply → OK and wait for installation. You do not need to install every Android API level; install the platform required by your Flutter project.

5. Create an Android emulator

• Open More Actions → Virtual Device Manager (or Device Manager inside a project).

• Choose Create Device, then select a phone model such as a Pixel device.

• Choose a recommended stable system image. Download it if needed, then click Next.

• Review the configuration and click Finish.

• Press the Play/Run icon beside the virtual device to start it.

If the emulator is slow or will not start, check that CPU virtualization is enabled, close other heavy apps, and consider using a physical Android phone with USB debugging.

6. Install the Flutter SDK

Android Studio is the IDE; Flutter is a separate SDK. Follow these steps if your goal is Flutter development.

• Open the official Windows Flutter guide: https://docs.flutter.dev/get-started/install/windows

• Download the stable Flutter SDK archive.

• Extract it to a short folder path you can write to, for example C:\src\flutter. Avoid protected folders such as Program Files.

• Add the Flutter bin folder to your Windows user PATH: search Windows for Edit environment variables → select Path → Edit → New → add C:\src\flutter\bin → save.

• Close and reopen Command Prompt or PowerShell so it reloads PATH.

7. Connect Flutter to Android Studio

• Open Android Studio → Settings → Plugins (on macOS, Preferences → Plugins).

• Search for Flutter and select Install. Install the Dart plugin if prompted.

• Restart Android Studio when asked.

• Open Command Prompt or the Android Studio terminal and run the following commands:

```
flutter doctor
flutter doctor --android-licenses
```

Read each flutter doctor section. Accept Android SDK licenses when prompted. Re-run flutter doctor afterwards to confirm the Android toolchain is ready.

8. Create and run a Flutter project

```
flutter create my_first_app
cd my_first_app
flutter run
```

• Start your emulator first, or connect a physical Android phone with Developer options and USB debugging enabled.

• In Android Studio, choose Open and select the project folder (the folder containing pubspec.yaml).

• Wait for project setup to finish, select the running device, and click Run.

9. Common problems and fixes

Problem	What to try
flutter is not recognized	Confirm the Flutter bin folder is in PATH; reopen the terminal.
Android licenses not accepted	Run flutter doctor --android-licenses and accept the prompts.
No devices found	Start the emulator or connect a phone; check adb devices.
Emulator does not start	Check virtualization and emulator system image; try a physical phone.
NDK missing source.properties	In SDK Manager → SDK Tools, repair/reinstall the specific NDK version required by the project.
Insufficient emulator storage	Wipe emulator data or edit the virtual device storage; back up anything needed first.

10. Final verification checklist

■ Android Studio opens successfully.

■ Android SDK Platform-Tools and Command-line Tools are installed.

■ A virtual device starts, or a physical phone is detected.

■ flutter --version prints a version (for Flutter projects).

■ flutter doctor reports no blocking Android toolchain issues.

■ A sample Flutter app runs on the selected device.

Important: Android Studio menus and recommended SDK versions change over time. Use the official Android and Flutter documentation linked above if a screen looks different.